



Lake surpassing 60 FPS in numerous popular gaming titles at 1080p medium settings, with some games like Counter Strike 2, DOTA 2, and GTA V exceeding 120 FPS at 1080p medium settings. Intel didn't hesitate to challenge Qualcomm's gaming performance, claiming a significant 68% lead over Qualcomm Snapdragon X-Elite powered laptops at the same resolution and settings. Additionally, Intel took aim at its rival AMD, asserting an average lead of 16% over AMD when comparing the AMD Ryzen HX 370-based laptop to an Intel Core Ultra 9 288V-powered laptop. While Intel's dominance over Qualcomm in gaming performance is unquestionable, the comparison with AMD Ryzen HX 370 is more nuanced, as both Intel and AMD will likely exchange blows depending on the games used for benchmarking. We eagerly await the opportunity to test the gaming capabilities of a Lunar Lake laptop.

Staggering battery life numbers

We checked out a few laptops at the demo zone and the common narrative across the show floor was "Highest battery life ever". OEMs were eager to highlight the impressive battery life of their devices, and the claimed numbers were nothing short of staggering. For instance, the Lunar Lake-powered Dell XPS 13 boasted an astonishing up to 26 hours of battery life. Intel's own rigorous benchmarking using the UL Procyon Office Productivity test claimed an impressive 20.1 hours of tested battery life for an Intel Core Ultra 7 268V-powered laptop, surpassing the 18.4

hours achieved by a Qualcomm Snapdragon X-Elite powered machine. Notably, Intel ensured a level playing field by using identical chassis and battery pack capacities for both devices.

While battery life claims are not uncommon, it's essential to consider the specific configurations, testing conditions, and workloads being used. However, Intel's demonstration of superior battery life across various aspects, including graphics, processor, and battery life, suggests a strong readiness to challenge Qualcomm and AMD in this critical area. Intel's consistent comparisons between Lunar Lake and Snapdragon Elite systems underscore its confidence in the platform's capabilities. With Lunar Lake, Intel seems poised to shatter the myth that X86 PCs can't deliver exceptional battery life.

AI performance

Before delving into the NPU improvement numbers, Intel emphasized the importance of a comprehensive AI PC framework. Intel asserts that while NPU performance is crucial, it's not the sole determining factor for all AI workloads. Many AI tasks rely on the CPU and, more significantly, the GPU for execution. We concur with Intel's stance on the vital role of a capable and power-efficient GPU, particularly in generative AI models where output time is more critical than power consumption. This explains Intel's focus on its graphics engine in Lunar Lake. According to Intel, 40% of AI application developers prioritize GPU optimization for AI tasks.

Lunar Lake-powered laptops can achieve up to 120 TOPS of combined AI

performance, with the GPU reaching 67 TOPS, followed by the NPU at up to 48 TOPS, and the CPU at up to 5 TOPS. While TOPS is currently the standard measurement metric for AI performance, Intel notes that it may not be the most optimal or realistic measure. As AI performance and measurement benchmarks evolve, TOPS may become less relevant. Intel compared Lunar Lake performance numbers across UL Procyon AI benchmarks and Geekbench ML benchmarks to demonstrate its lead over Qualcomm and AMD. We look forward to testing our review unit to determine the true leader in AI benchmarks. By adopting a holistic approach to AI performance, Intel is poised to revolutionize the industry's understanding of AI capabilities.

Shots fired!

The story is clear: Intel is pulling out all the stops to regain its crown in the space, particularly in terms of efficiency and AI performance. The goal is to shatter the myth that X86 PCs are incapable of delivering excellent battery life. If the numbers are to be believed, they may be well on their way to achieving this. However, this is not just a battle between Qualcomm or AMD and Intel; it's a resurgence of the classic ARM vs X86 rivalry. We saw AMD emphasize performance during the launch of the AMD Ryzen AI 300 series in July, and now Intel is seeking to lead and define the narrative around power efficiency and performance per watt across CPU, GPU, and NPU. We will reserve our final judgment until we receive our review units based on the Intel Lunar Lake platform. **[I]**